

Performance Test

Test Run ID: 90fe43c9-4d1d-4ec8-8499-492580ff1125

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	05 June 2026	Max VUs Tested	500
Executor	ramping-arrival-rate	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a load test, utilizing a ramping arrival rate scenario to simulate a gradual increase in traffic. The load profile consists of a maximum of 500 Virtual Users (VUs) with a pre-allocated capacity of 500 VUs, ramping up to a peak of 250 requests per second (RPS) over a 30-second duration.

Stable Threshold	Degradation Onset	Critical Point	Peak Throughput
64-250 TPS	N/A	N/A	250 TPS

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	264.36ms	2319.13ms	233.34ms	213.34ms	357.84ms	423.32ms	-
http_req_waiting	262.28ms	1599.43ms	233.02ms	213.18ms	356.61ms	417.71ms	-
http_req_connecting	32.22ms	495.74ms	0.00ms	0.00ms	224.55ms	243.00ms	-
iteration_duration	1328.36ms	3320.08ms	1235.72ms	1213.98ms	1683.57ms	1758.81ms	-

Counters

metric	count	rate
data_received	16.47 MB	531.29 kB/s
data_sent	1.05 MB	33.99 kB/s
http_reqs	4,499	141.69/s
iterations	4,499	141.69/s

Rates

metric	rate
checks_succeeded	100.00% (4499/4499)
checks_failed	0.00% (0/4499)
http_req_failed	0.00% (0/4499)

Gauges

metric	value
vus	106 (min: 51, max: 307)
vus_max	500 (min: 500, max: 500)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	4499	0	100.00%	Pass

3. Advanced Analysis

3.1 Error Analysis

- No error or warning logs found in the test execution.
- No metrics abnormalities, error rate spikes, or failed checks detected.

3.2 Observation

The performance test results indicate a stable and successful execution of the test. However, upon closer inspection, there are some subtle patterns worth noting.

- The `http\_req\_duration` metric shows a high average duration of 264.364ms, with a maximum value of 2319.133ms. This suggests that there may be some latency issues in the system, particularly for certain requests.
- The `http\_req\_connecting` and `http\_req\_tls\_handshaking` metrics show significant average and maximum values, indicating that there may be some issues with connection establishment and TLS handshaking.
- The `http\_req\_sending` metric shows a relatively low average value, suggesting that the system is able to send requests efficiently.
- The `http\_req\_receiving` metric shows a relatively high average value, suggesting that the system is experiencing some issues with receiving responses.

These patterns suggest that the system may be experiencing some latency and connection issues, particularly for certain requests. Further investigation is needed to determine the root cause of these issues.

3.3 Final Recommendations

1. **Advanced Connection Management:** Implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.
2. **Session Persistence sticky affinity/Redis cache:** Implement a session persistence mechanism to reduce the overhead of creating new sessions for each request.
3. **Gateway resilience Istio outlier circuit breaker:** Implement an outlier detection mechanism to detect and prevent cascading failures in the system.
4. **Observability tracing and standardized log levels:** Implement a tracing mechanism to provide detailed information about request flow and latency, and standardize log levels to improve debugging and troubleshooting.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement a read/write splitting mechanism to reduce the load on the database, and use a connection pooling mechanism to reduce the overhead of establishing new connections to the database.

3.4 Conclusion

The performance test results indicate a stable and successful execution of the test, but with some subtle patterns worth noting. Further investigation is needed to determine the root cause of these issues. Based on the analysis, the following recommendations have been made to improve the performance and resilience of the system.